Automatic Emergency Braking (AEB) System Architecture Document

This document presents the detailed architecture, signal flow, subsystem interaction, and implementation methodology of the Automatic Emergency Braking (AEB) system developed using MATLAB/Simulink integrated with IPG CarMaker.

---

1. System Objective

The primary objective of the AEB system is to detect collision threats ahead of the ego vehicle and automatically generate controlled braking commands to prevent or mitigate collisions.

2. System Architecture Overview

Subsystem	Primary Function	Key Signals
Radar Processing	Distance and velocity estimation	DistX, VrelX
Camera Processing	Object validation and classification	ObjID, Type, Confidence
Decision Logic	Collision assessment and TTC estimation	TTC, Safe Distance
Brake Controller	Brake demand generation	BrakeCmd
Vehicle Interface	Brake application to ego vehicle	VhclCtrl.Brake

3. Radar Sensor Architecture

The radar subsystem acts as the primary perception module for longitudinal collision estimation. The radar provides stable distance and relative velocity measurements required for TTC estimation.

Signal	Description	Purpose
DistX	Longitudinal target distance	Collision estimation
VrelX	Relative longitudinal velocity	Closing speed estimation
ProbExist	Probability of target existence	Detection validation
ProbDetect	Probability of target detection	Sensor reliability

4. Camera Sensor Architecture

The camera subsystem is utilized primarily for environmental validation and object classification. The camera supports object confirmation during AEB operation.

Signal	Description	Usage
ObjID	Detected object identifier	Object validation
Type	Object classification	Vehicle/Pedestrian recognition
Confidence	Detection confidence	Detection filtering
nVisPixels	Visible object pixels	Relative proximity estimation

5. Collision Assessment Logic

The collision assessment subsystem computes Time-To-Collision (TTC) using distance and relative velocity. Braking logic is activated when TTC falls below the predefined threshold.

Parameter	Typical Value
Safe Stopping Distance	3 m
TTC Threshold	1.5 s
Operating Speed	30 km/h

6. Brake Control Strategy

Smooth braking behavior is achieved using saturation blocks, gain tuning, transfer function smoothing, and velocity feedback mechanisms. The brake controller prevents oscillatory stop-start behavior.

7. Simulink Implementation

The implementation uses Simulink subsystems and CarMaker sensor interfaces. Bus selectors are used for extracting sensor data, while transfer functions and saturation blocks ensure smooth longitudinal control.

Simulink Block	Purpose
Radar Sensor	Front radar acquisition
Radar Sensor Object	Object-level radar data
Bus Selector	Signal extraction
Gain	Brake scaling
Transfer Function	Brake smoothing
Saturation	Brake limit enforcement

Switch	Conditional brake logic

8. Conclusion

The implemented AEB architecture provides reliable collision detection and smooth emergency braking behavior suitable for simulation-based validation in IPG CarMaker environments.